

# RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

## 4. RF/Microwave Components and Devices

8 hrs · asked in **all 24 papers** · 11 syllabus sub-topics · **411 marks, 19.8% of the archive, 65 questions** · **124 numerical (30%) + 287 theory** → one derivation (the waveguide) worth up to 10 marks, one device (the magic tee) asked 9 times by name and 8 more in disguise, and a ring of 4 to 6 mark short notes.

### READ THIS FIRST

Chapter 4 has the biggest question count in the paper, but three scripts answer most of it:

- Field equations of a rectangular waveguide, TE or TM** (14 papers, up to **10 marks**) → §4.1. **One seven-step script covers both modes**: only the boundary condition in Step 5 changes. Learn TM first; TE is the same page with sin and cos swapped.
- Mode questions** (9 papers, 1 to 5 marks each): dominant, degenerate, “prove  $TM_{01}$  cannot exist”, “show  $TM_{11}$  is the lowest TM mode” → §4.2. Every one is **the cut-off formula plus one sentence**.
- Magic tee** (9 papers) → §4.5, construction, derivation and matrix. The same matrix answers **identify the device from its S-matrix** (4 papers, §4.6) and the **radar power problems** (4 papers, §4.7): **one matrix, three families**. The combinations (two tees joined, arms terminated or shorted) are worked in the numerical companion.

Two more things worth knowing before the paper starts:

- Circular waveguide** (4 papers, 7 to 10 marks) is the rectangular script again in cylindrical coordinates, with Bessel functions in place of sines. The only new fact to memorise is the **root table** of §4.3.
- Short notes** rotate among circulator (3 papers), directional coupler (3), cavity resonator (2), Gunn diode, MASER, probe/loop coupling and microstrip (1 each). Each gets a ready answer below.

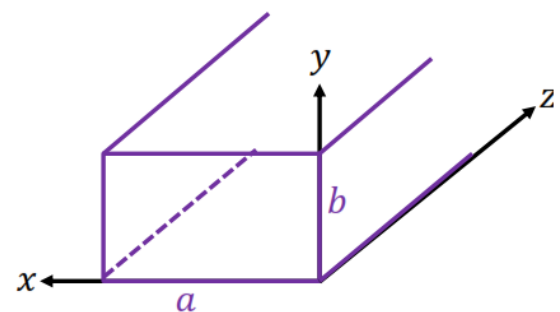
### 4.1 Rectangular Waveguide: TE and TM Field Equations

Derive the field equations and characteristic parameters of a rectangular waveguide operating in TE / TM mode **4 5 7+3 8 8+2 10**

**TOP 14/24** 69 Bh, 70 Bh, 70 Ma, 71 Bh, 72 Ma, 74 Ma, 74 Bh, 76 Bh, 77 Ch, 79 Ch, 80 Ch, 80 Bh, 81 Ba, 81 Bh · **the most repeated ask in the chapter: TE 6 papers, TM 7, modes in general 1**

#### Set-up

- Hollow metal pipe, inner broad wall  $a$  along  $x$ , narrow wall  $b$  along  $y$ , propagation along  $+z$ ,  $a > b$ .
- Lossless dielectric ( $\mu$ ,  $\epsilon$ ) inside, perfectly conducting walls.
- Symbols follow the **RF guide** (*Fundamentals of Microwave Engineering*, §4.1.1, eqs 4.1 to 4.37, Liao's scheme):
  - $\gamma$ : intrinsic propagation constant of the dielectric; lossless  $\Rightarrow \gamma = j\omega\sqrt{\mu\epsilon}$ ,  $\gamma^2 = -\omega^2\mu\epsilon$ .
  - $\gamma_g = \alpha_g + j\beta_g$ : propagation constant in the guide.
  - $k_x, k_y$ : separation constants;  $k_c = \sqrt{k_x^2 + k_y^2}$ , cut-off wave number.
  - $\psi$ : stands for  $H_z$  (TE) or  $E_z$  (TM).
- No TEM mode**. With  $E_z = H_z = 0$  every transverse component in Step 4 is zero. Physically, a closed  $H$  loop in the cross-section needs an axial current, and a single hollow conductor has no centre conductor to carry it.



Co-ordinates of the rectangular guide. Source: Gangaju, Ch-4 deck, slide 12.

**Step 1: wave equations (from Maxwell's equations)**

- Vector wave equations:  $\nabla^2 E = \gamma^2 E$ ,  $\nabla^2 H = \gamma^2 H$ ,  $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$ .
- Each rectangular component  $\psi$  obeys the scalar Helmholtz equation  $\nabla^2 \psi = \gamma^2 \psi$ , i.e.  $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \gamma^2 \psi$ .

### Step 2: separation of variables

- Put  $\psi = X(x)Y(y)Z(z)$ , substitute, divide by  $XYZ$ :  $\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = \gamma^2$ .
- Each term is a function of one variable only  $\Rightarrow$  each is a constant:
  - $\frac{d^2 X}{dx^2} = -k_x^2 X \Rightarrow X = A \sin k_x x + B \cos k_x x$ .
  - $\frac{d^2 Y}{dy^2} = -k_y^2 Y \Rightarrow Y = C \sin k_y y + D \cos k_y y$ .
  - $\frac{d^2 Z}{dz^2} = \gamma_g^2 Z \Rightarrow Z = e^{-\gamma_g z}$  (wave along  $+z$  only).
- Adding the constants back:  $-k_x^2 - k_y^2 + \gamma_g^2 = \gamma^2$ , so  $\boxed{\gamma_g^2 = \gamma^2 + k_x^2 + k_y^2 = \gamma^2 + k_c^2}$ .
- Lossless dielectric,  $\gamma^2 = -\omega^2 \mu \epsilon$ :  $\boxed{\gamma_g = \pm \sqrt{k_c^2 - \omega^2 \mu \epsilon}}$ .

### Step 3: three cases for $\gamma_g$

- **Case I**,  $\omega^2 \mu \epsilon = k_c^2$ :  $\gamma_g = 0$ , **cut-off**:  $f_c = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{k_x^2 + k_y^2}$ .
- **Case II**,  $\omega^2 \mu \epsilon > k_c^2$ :  $\gamma_g = \pm j\beta_g = \pm j\omega\sqrt{\mu\epsilon}\sqrt{1 - (f_c/f)^2}$ , imaginary  $\rightarrow$  **propagates**; needs  $f > f_c$ .
- **Case III**,  $\omega^2 \mu \epsilon < k_c^2$ :  $\gamma_g = \pm \alpha_g = \pm \omega\sqrt{\mu\epsilon}\sqrt{(f_c/f)^2 - 1}$ , real  $\rightarrow$  **attenuated** as  $e^{-\alpha_g z}$ , carries no power;  $f < f_c$ . A waveguide is a **high-pass filter**.
- Final solution (propagating wave):

$$\psi = [A \sin k_x x + B \cos k_x x] [C \sin k_y y + D \cos k_y y] e^{-j\beta_g z}$$

### Step 4: transverse components from $E_z$ and $H_z$ (expand the curl equations)

- Every component carries  $e^{-j\beta_g z}$ , so  $\partial/\partial z \rightarrow -j\beta_g$ . Expand  $\nabla \times E = -j\omega\mu H$  and  $\nabla \times H = j\omega\epsilon E$  in components and solve the four transverse equations:

$$\begin{aligned} E_x &= -\frac{j\beta_g}{k_c^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial y}, & E_y &= -\frac{j\beta_g}{k_c^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial x}, \\ H_x &= -\frac{j\beta_g}{k_c^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial y}, & H_y &= -\frac{j\beta_g}{k_c^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial x}. \end{aligned}$$

- **This block is the whole trick**: find  $H_z$  (TE) or  $E_z$  (TM) and the other four components follow by differentiation. The guide prints only the TE half (4.22) and the TM half (4.28 to 4.31); this is both at once.

### Step 5 (TE): $E_z = 0$ , $H_z \neq 0$ , so $\psi = H_z$

- With  $E_z = 0$ :  $E_x \propto \partial H_z / \partial y$ ,  $E_y \propto \partial H_z / \partial x$ ; tangential  $E$  vanishes on the walls.
- $x = 0$ :  $E_y = 0 \Rightarrow \partial H_z / \partial x = 0 \Rightarrow A = 0$ .
- $y = 0$ :  $E_x = 0 \Rightarrow \partial H_z / \partial y = 0 \Rightarrow C = 0$ .
- $x = a$ :  $\sin k_x a = 0 \Rightarrow k_x = m\pi/a$ .
- $y = b$ :  $\sin k_y b = 0 \Rightarrow k_y = n\pi/b$ .

$$H_z = H_{0z} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta_g z}$$

$$H_{0z} = BD; m, n \geq 0, \text{ not both zero.}$$

### Step 5 (TM): $H_z = 0$ , $E_z \neq 0$ , so $\psi = E_z$

- $E_z$  is tangential to every wall, so it must vanish there.
- $x = 0$ :  $E_z = 0 \Rightarrow B = 0$ .
- $y = 0$ :  $E_z = 0 \Rightarrow D = 0$ .
- $x = a$ :  $\sin k_x a = 0 \Rightarrow k_x = m\pi/a$ .
- $y = b$ :  $\sin k_y b = 0 \Rightarrow k_y = n\pi/b$ .

$$E_z = E_{0z} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta_g z}$$

$$E_{0z} = AC; m, n \geq 1; m = 0 \text{ or } n = 0 \text{ kills every component.}$$

**Step 6: substitute into Step 4 (write  $e^{-j\beta_g z}$  on every line in the exam)****TE<sub>mn</sub>** ( $E_z = 0$ )

$$\begin{aligned}
H_x &= -\frac{j\beta_g}{k_c^2} \frac{\partial H_z}{\partial x} = H_{0x} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta_g z}, & H_{0x} &= \frac{j\beta_g}{k_c^2} \frac{m\pi}{a} H_{0z} \\
H_y &= -\frac{j\beta_g}{k_c^2} \frac{\partial H_z}{\partial y} = H_{0y} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta_g z}, & H_{0y} &= \frac{j\beta_g}{k_c^2} \frac{n\pi}{b} H_{0z} \\
E_x &= -\frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial y} = E_{0x} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta_g z}, & E_{0x} &= \frac{j\omega\mu}{k_c^2} \frac{n\pi}{b} H_{0z} \\
E_y &= \frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial x} = E_{0y} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta_g z}, & E_{0y} &= -\frac{j\omega\mu}{k_c^2} \frac{m\pi}{a} H_{0z}
\end{aligned}$$

**TM<sub>mn</sub>** ( $H_z = 0$ )

$$\begin{aligned}
E_x &= -\frac{j\beta_g}{k_c^2} \frac{\partial E_z}{\partial x} = E_{0x} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta_g z}, & E_{0x} &= -\frac{j\beta_g}{k_c^2} \frac{m\pi}{a} E_{0z} \\
E_y &= -\frac{j\beta_g}{k_c^2} \frac{\partial E_z}{\partial y} = E_{0y} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta_g z}, & E_{0y} &= -\frac{j\beta_g}{k_c^2} \frac{n\pi}{b} E_{0z} \\
H_x &= \frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial y} = H_{0x} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta_g z}, & H_{0x} &= \frac{j\omega\epsilon}{k_c^2} \frac{n\pi}{b} E_{0z} \\
H_y &= -\frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial x} = H_{0y} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta_g z}, & H_{0y} &= -\frac{j\omega\epsilon}{k_c^2} \frac{m\pi}{a} E_{0z}
\end{aligned}$$

All eight components are checked in `src/wg.py` against both curl equations and against tangential  $E = 0$  on all four walls; a single flipped sign fails the check.

**Step 7: the characteristic equations (the “+2” or “+3” of every version)**

| Quantity                | Expression                                                                                                 | Note                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Cut-off wave number     | $k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 = \omega_c^2 \mu \epsilon$          | TE and TM alike                                                           |
| Cut-off frequency       | $f_c = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$       | from $k_c = \omega_c \sqrt{\mu\epsilon}$ ; air: $\frac{c}{2}\sqrt{\cdot}$ |
| Cut-off wavelength      | $\lambda_c = \frac{2\pi}{k_c} = \frac{1}{\sqrt{(m/2a)^2 + (n/2b)^2}}$                                      | TE <sub>10</sub> : $\lambda_c = 2a$                                       |
| Phase constant          | $\beta_g = \sqrt{\omega^2 \mu \epsilon - k_c^2} = \omega \sqrt{\mu\epsilon} \sqrt{1 - (f_c/f)^2}$          | above cut-off                                                             |
| Guide wavelength        | $\lambda_g = \frac{2\pi}{\beta_g} = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$                                  | $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$     |
| Phase velocity in guide | $v_g = \frac{\omega}{\beta_g} = \frac{v_p}{\sqrt{1 - (f_c/f)^2}}$                                          | $> v_p$ ; guide's name, see below                                         |
| Group velocity          | $v_{gr} = \frac{d\omega}{d\beta_g} = v_p \sqrt{1 - (f_c/f)^2}$                                             | $< v_p$ ; $v_g v_{gr} = v_p^2$                                            |
| Wave impedance, TE      | $Z_g = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\omega\mu}{\beta_g} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$ | $> \eta$                                                                  |
| Wave impedance, TM      | $Z_g = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\beta_g}{\omega\epsilon} = \eta \sqrt{1 - (f_c/f)^2}$    | $< \eta$                                                                  |

- $v_p = 1/\sqrt{\mu\epsilon}$  is the phase velocity in the unbounded dielectric,  $\eta = \sqrt{\mu/\epsilon}$  the intrinsic impedance; for air  $v_p = c$  and  $\eta_0 = 120\pi \approx 377 \Omega$ .
- **Name clash:** the guide's  $v_g$  is the phase velocity **in the guide** (Liao's name). The companion and most books use  $v_p$  for that and  $v_g$  for group velocity. Say which one you mean in the exam.
- Other decks: Gangaju writes  $h$  for  $k_c$  and  $\gamma$  for  $\beta_g$ ; Pozar writes  $k_c, \beta, Z_{TE} / Z_{TM}$ .

**RF guide misprints (book pp. 32 to 36), corrected above**

- (4.1)  $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$ : the  $j$  is missing,  $\sigma + j\omega\epsilon$ .
- (4.6) prints  $\frac{1}{X}$  in front of all three terms: it is  $\frac{1}{X}$ ,  $\frac{1}{Y}$ ,  $\frac{1}{Z}$ .
- (4.9), (4.12)  $Z'' = -k_z^2 Z$  with a sin/cos solution contradict its own  $\gamma_g$  in (4.14) and  $e^{-j\beta_g z}$  in (4.19):  $Z'' = \gamma_g^2 Z$ ,  $Z = e^{-\gamma_g z}$ .
- (4.15)  $\gamma_g = \pm\sqrt{\omega_c^2\mu\epsilon - k_c^2}$ : root inverted and  $\omega_c$  for  $\omega$ ; it is  $\pm\sqrt{k_c^2 - \omega^2\mu\epsilon}$ . Cases II and III also print  $\omega_c$  where  $\omega$  is meant.
- (4.18)  $\gamma_g = \pm j\alpha_g$ : attenuation is real,  $\gamma_g = \pm\alpha_g$ .
- (4.19) prints sin in all four places: second term of each bracket is cos.
- (4.22)  $E_y = -\frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial y}$ : two errors, it is  $+\frac{j\omega\mu}{k_c^2} \frac{\partial H_z}{\partial x}$ . TE boundary conditions are skipped entirely; Step 5 supplies them.
- (4.24), (4.35)  $\lambda_g = 1/\sqrt{1 - (f_c/f)^2}$ : numerator  $\lambda$  is missing.
- (4.26)  $Z_g = E_x/E_y$ : it is  $E_x/H_y$ .
- (4.29) TM  $E_y$  amplitude labelled  $H_{0y}$ : it is  $E_{0y}$ .
- (4.31)  $H_y = +\frac{j\omega\epsilon}{k_c^2} \frac{\partial E_z}{\partial x}$ : sign is  $-$ .

Gangaju deck slide 18 prints  $k_y = n\pi/a$ ; it is  $n\pi/b$ .

**Answer skeleton for the 4 to 5 mark short-note version** 4 5

- Figure of the guide with  $a, b, z$  (1 line).
- Helmholtz equation,  $\psi = XYZ$ , and  $\gamma_g^2 = \gamma^2 + k_c^2$  in two lines.
- The one boundary-condition result ( $H_z = H_{0z} \cos \cos$  for TE,  $E_z = E_{0z} \sin \sin$  for TM).
- The four transverse components from Step 6, without re-deriving them.
- $f_c$  and  $Z_g$ . For the 10-mark version add Steps 2 to 5 in full and the whole Step 7 table.

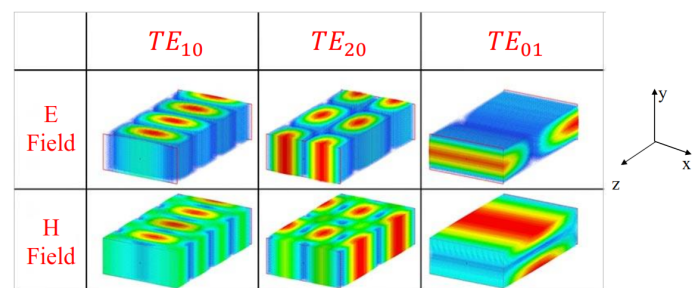
**4.2 Modes: Dominant, Degenerate, and the Short Proofs****Dominant and degenerate modes; show which modes cannot exist; compare modes by cut-off** 1+1+2

2 2+2 3 5

**TOP 9/24** 70 Bh, 70 Ma, 71 Bh, 72 Ma, 72 Ash, 80 Bh, 81 Bh, 82 Ba, 82 Bh · every one is the  $f_c$  formula plus one sentence

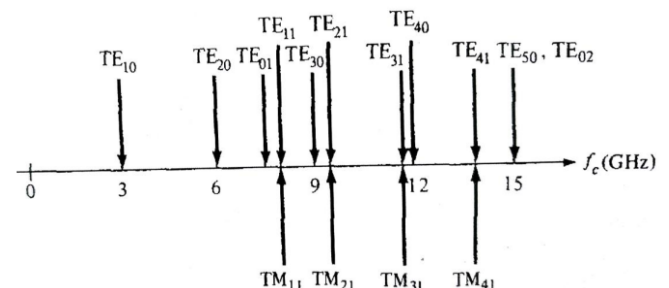
**Definitions**

- **Mode**: one allowed field pattern, labelled by the integers  $m, n$  = number of half-cycle variations across  $a$  and across  $b$ .
- **Dominant mode**: the mode with the **lowest cut-off frequency**.
  - Rectangular guide ( $a > b$ ): **TE<sub>10</sub>**,  $f_c = c/2a$ ,  $\lambda_c = 2a$ .
  - Circular guide: **TE<sub>11</sub>**,  $f_c = 1.841 c/2\pi r$  (§4.3).
  - Between  $f_{c,10}$  and the next cut-off only the dominant mode propagates → **single-mode** operation, the reason guides are sized this way.
- **Degenerate modes**: different field patterns with the **same cut-off frequency**.
  - TE <sub>$m$  $n$</sub>  and TM <sub>$m$  $n$</sub>  for every  $m, n \geq 1$ : same  $f_c$  formula.
  - Square guide ( $a = b$ ): TE<sub>10</sub> and TE<sub>01</sub> as well.
  - Circular guide: TE<sub>01</sub> and TM<sub>11</sub> (both root 3.832).
  - Unwanted, because power can couple between them at any discontinuity.



E and H field of TE<sub>10</sub>, TE<sub>20</sub>, TE<sub>01</sub>: one, two and one half-cycles across the walls.

Source: Gangaju, slide 20.



Cut-off ladder, Gangaju slide 22 (from Sadiku). The slide's caption says  $a = 2.5$  cm,  $b = 1$  cm; the arrows (TE<sub>10</sub> at 3 GHz, TE<sub>20</sub> at 6, TE<sub>50</sub> and TE<sub>02</sub> at 15) are those of a 5 cm × 2 cm guide. TM arrows below the line sit on their TE twins: degeneracy.

verified/added

**Prove that  $TM_{01}$  and  $TM_{10}$  modes do not exist** 3

70 Ma

- TM:  $H_z = 0$  and  $E_z = E_{0z} \sin(m\pi x/a) \sin(n\pi y/b) e^{-j\beta_g z}$ .
- $m = 0$  or  $n = 0 \Rightarrow \sin 0 = 0 \Rightarrow E_z = 0$  everywhere.
- Every transverse component is a derivative of  $E_z$  (Step 4 with  $H_z = 0$ )  $\Rightarrow E_x = E_y = H_x = H_y = 0$ .
- All six components vanish  $\Rightarrow$  **no field, no mode**.  $TM_{mn}$  needs  $m \geq 1$  and  $n \geq 1$ .

**Prove that the lowest cut-off TM mode is  $TM_{11}$**  3

82 Bh

- From above, TM needs  $m, n \geq 1$ .
- $f_c = \frac{c}{2} \sqrt{(m/a)^2 + (n/b)^2}$  grows with both  $m$  and  $n$ .
- Smallest admissible pair is  $m = n = 1$ :  $f_{c, TM_{11}} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$ , for any  $a, b$ .

**Show that the dominant TE mode is  $TE_{10}$ ; compare  $TE_{10}$  with  $TE_{20}$**  2 3

71 Bh, 81 Bh

- TE:  $H_z = H_{0z} \cos \cos$  survives with one index zero ( $\cos 0 = 1$ ); only  $TE_{00}$  is excluded (constant  $H_z$ , all transverse components zero).
- Two candidates for the lowest  $f_c$ :  $f_{c,10} = \frac{c}{2a}$ ,  
 $f_{c,01} = \frac{c}{2b}$ .
- $a > b \Rightarrow f_{c,10} < f_{c,01} \Rightarrow$   **$TE_{10}$  is dominant**. It also beats  $TM_{11}$ , since  $\frac{1}{a^2} < \frac{1}{a^2} + \frac{1}{b^2}$ .
- **81 Bh prints "when  $b > a$ ".** With  $b > a$  it is  $TE_{01}$  that is dominant;  $TE_{10}$  needs  $a > b$ . The paper has the inequality reversed. Prove the  $a > b$  case and state this.

verified/added

|                      | <b><math>TE_{10}</math></b> | <b><math>TE_{20}</math></b> |
|----------------------|-----------------------------|-----------------------------|
| $f_c$                | $c/2a$                      | $c/a = 2f_{c,10}$           |
| $\lambda_c$          | $2a$                        | $a$                         |
| Variation across $a$ | one half-cycle              | two half-cycles             |
| Field components     | $E_y, H_x, H_z$ only        | same three                  |
| Status               | <b>dominant</b>             | higher-order                |

- Single-mode band:  $c/2a < f < c/a$  provided  $a \geq 2b$  (otherwise  $TE_{01}$  arrives first at  $c/2b$ ).

**Dominant vs degenerate modes; for  $a = 3b$  find the dominant mode among  $TM_{01}, TM_{10}, TM_{11}, TM_{21}, TM_{12}, TM_{02}, TM_{20}$** 

2+2 3

72 Ma, 82 Ba · full working in the companion, Problem 4

- With  $a = 3b$ :  $f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$ .
- $TM_{01}, TM_{10}, TM_{02}, TM_{20}$  **do not exist** (an index is zero).
- $TM_{11}$ :  $\sqrt{10} = 3.162$ ;  $TM_{21}$ :  $\sqrt{13} = 3.606$ ;  $TM_{12}$ :  $\sqrt{37} = 6.083$  (in units of  $c/2a$ ).
- **Dominant among those listed:  $TM_{11}$** . (Overall,  $TE_{10}$  at 1 unit.)

**Cut-off checks: does the guide pass the signal?** 1+1+2 3 5

70 Bh, 72 Ash, 80 Bh · worked in the companion, Problem 3

- Rule: compute  $f_c$  of the **dominant** mode; the signal propagates only if  $f > f_c$ .
- **70 Bh**: wall separation 5 cm,  $TE_{10}$ :  $f_c = 3 \text{ GHz} > 1 \text{ GHz} \Rightarrow$  **1 GHz cannot propagate**; it decays at 59.2 Np/m.
- **80 Bh**: width 2.254 cm:  $f_c = 6.65 \text{ GHz} > 6 \text{ GHz} \Rightarrow$  **6 GHz is not supported**.

**4.3 Circular Waveguide****Field equations of a circular waveguide in TE / TM mode; TE vs TM; dominant mode** 7 8+2 10

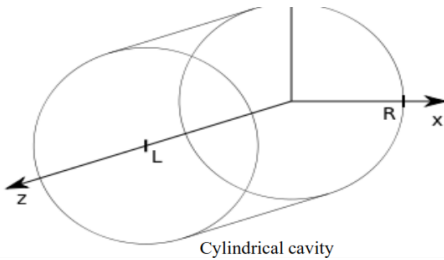
HOT 4/24 71 Ma, 75 Bh, 79 Bh, 82 Ba · the rectangular script in cylindrical co-ordinates

**Set-up**

- Hollow circular pipe of inner radius  $a$ , co-ordinates  $(\rho, \phi, z)$ , fields  $\propto e^{-j\beta z}$ ;  $k_c^2 = k^2 - \beta^2$ , with  $k = \omega\sqrt{\mu\epsilon}$ .
- **Step 1**, the cylindrical version of the transverse equations (Waveguide.pdf p8, Pozar §3.4):

$$E_\rho = -\frac{j}{k_c^2} \left( \beta \frac{\partial E_z}{\partial \rho} + \frac{\omega \mu}{\rho} \frac{\partial H_z}{\partial \phi} \right), \quad E_\phi = -\frac{j}{k_c^2} \left( \frac{\beta}{\rho} \frac{\partial E_z}{\partial \phi} - \omega \mu \frac{\partial H_z}{\partial \rho} \right),$$

$$H_\rho = \frac{j}{k_c^2} \left( \frac{\omega \epsilon}{\rho} \frac{\partial E_z}{\partial \phi} - \beta \frac{\partial H_z}{\partial \rho} \right), \quad H_\phi = -\frac{j}{k_c^2} \left( \omega \epsilon \frac{\partial E_z}{\partial \rho} + \frac{\beta}{\rho} \frac{\partial H_z}{\partial \phi} \right).$$



Cylindrical cavity

Cylindrical co-ordinates. The figure is a circular cavity (a guide closed at both ends, §4.10); the guide itself is the same cylinder, open along  $z$ . Source: Gangaju, slide 93.

**TE<sub>nm</sub>:**  $E_z = 0$

- $H_z = (A \sin n\phi + B \cos n\phi) J_n(k_c \rho) e^{-j\beta z}$ .
- **Boundary:**  $E_\phi(\rho = a) = 0$ . Since  $E_\phi \propto \partial H_z / \partial \rho$ :  
 $J'_n(k_c a) = 0 \Rightarrow \boxed{k_c = p'_{nm}/a}$ ,  $p'_{nm} = m$ -th root of  $J'_n$ .

$$E_\rho = \frac{-j\omega\mu n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$E_\phi = \frac{j\omega\mu}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$H_\rho = \frac{-j\beta}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$H_\phi = \frac{-j\beta n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$- f_c = \frac{p'_{nm}}{2\pi a \sqrt{\mu\epsilon}}, \quad Z_{TE} = \frac{E_\rho}{H_\phi} = \frac{\omega\mu}{\beta} = \frac{\eta k}{\beta}.$$

Root table and mode order (all a circular answer needs)

| Mode             | Root              | $\lambda_c$ | Order                            |
|------------------|-------------------|-------------|----------------------------------|
| TE <sub>11</sub> | $p'_{11} = 1.841$ | $3.41a$     | <b>1st, dominant</b>             |
| TM <sub>01</sub> | $p_{01} = 2.405$  | $2.61a$     | 2nd, lowest TM                   |
| TE <sub>21</sub> | $p'_{21} = 3.054$ | $2.06a$     | 3rd                              |
| TE <sub>01</sub> | $p'_{01} = 3.832$ | $1.64a$     | 4th (low-loss mode)              |
| TM <sub>11</sub> | $p_{11} = 3.832$  | $1.64a$     | degenerate with TE <sub>01</sub> |

$\lambda_c = 2\pi a / p$ . Roots from `scipy` in `wg.py`. Worked example:  $a = 2$  cm gives TE<sub>11</sub> at 4.40 GHz and TM<sub>01</sub> at 5.74 GHz (Waveguide.pdf p13).

**Step 2 and 3: separate the Helmholtz equation**

$$- \frac{\partial^2 h_z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial h_z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 h_z}{\partial \phi^2} + k_c^2 h_z = 0.$$

$$- h_z = R(\rho)P(\phi):$$

- $P'' + n^2 P = 0 \Rightarrow P = A \sin n\phi + B \cos n\phi$ ;  $n$  an integer because the field repeats every  $2\pi$ .

$$- \rho^2 R'' + \rho R' + (k_c^2 \rho^2 - n^2) R = 0: \text{Bessel's equation}$$

$$\Rightarrow R = C J_n(k_c \rho) + D Y_n(k_c \rho).$$

$$- Y_n \rightarrow \infty \text{ at } \rho = 0 \Rightarrow D = 0.$$

**TM<sub>nm</sub>:**  $H_z = 0$

$$- E_z = (A \sin n\phi + B \cos n\phi) J_n(k_c \rho) e^{-j\beta z}.$$

- **Boundary:**  $E_z(\rho = a) = 0$  directly:

$$J_n(k_c a) = 0 \Rightarrow \boxed{k_c = p_{nm}/a}, \quad p_{nm} = m\text{-th root of } J_n.$$

$$E_\rho = \frac{-j\beta}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$E_\phi = \frac{-j\beta n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$H_\rho = \frac{j\omega\epsilon n}{k_c^2 \rho} (A \cos n\phi - B \sin n\phi) J_n(k_c \rho)$$

$$H_\phi = \frac{-j\omega\epsilon}{k_c} (A \sin n\phi + B \cos n\phi) J'_n(k_c \rho)$$

$$- f_c = \frac{p_{nm}}{2\pi a \sqrt{\mu\epsilon}}, \quad Z_{TM} = \frac{E_\rho}{H_\phi} = \frac{\beta}{\omega\epsilon} = \frac{\eta\beta}{k}.$$

- **Sign:** Waveguide.pdf p11 prints  $E_\phi$  with a +; it is -. `wg.py` fails the curl equation with the +. verified/added

**TE vs TM in a circular guide (71 Ma)**

|              | TE <sub>nm</sub>                       | TM <sub>nm</sub>                            |
|--------------|----------------------------------------|---------------------------------------------|
| Longitudinal | $E_z = 0, H_z \neq 0$                  | $H_z = 0, E_z \neq 0$                       |
| Boundary     | $J'_n(k_c a) = 0$                      | $J_n(k_c a) = 0$                            |
| Roots        | $p'_{nm}$                              | $p_{nm}$                                    |
| Dominant     | TE <sub>11</sub> , 1.841               | TM <sub>01</sub> , 2.405                    |
| Impedance    | $\eta k / \beta > \eta$                | $\eta \beta / k < \eta$                     |
| $m = 0$ ?    | TE <sub>n0</sub> absent                | TM <sub>n0</sub> absent                     |
| Special      | TE <sub>01</sub> : loss falls with $f$ | TM <sub>01</sub> : symmetric, rotary joints |

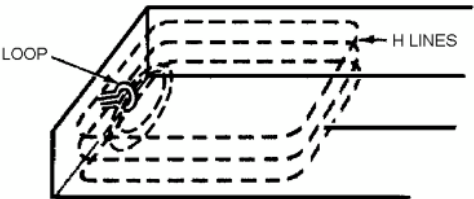
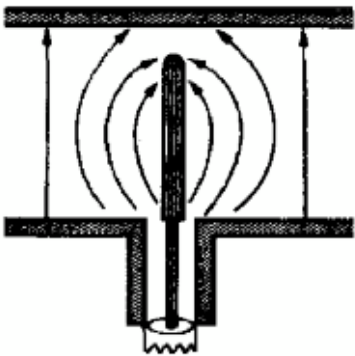
- **75 Bh:** dominant TE mode = TE<sub>11</sub>, dominant TM mode = TM<sub>01</sub>, and TE<sub>11</sub> is dominant overall since  $1.841 < 2.405$ .

4.4 Coupling Energy into a Waveguide: Probes, Loops and Slots

Probe coupling vs loop coupling 5

PIN 1/24 69 Bh · also the answer to “how is a cavity or klystron excited”

|               | Probe                                                                                          | Loop                                                                        |
|---------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Couples to    | E field                                                                                        | H field                                                                     |
| Acts as       | quarter-wave monopole antenna                                                                  | small magnetic dipole                                                       |
| Best position | centre of broad wall $a$ , parallel to E lines, $\lambda_g/4$ from the shorted end (E maximum) | where H is strongest, e.g. at the shorted end; loop plane normal to H lines |
| Less coupling | shorten, move off centre, shield                                                               | rotate, or move to fewer H lines                                            |
| Bandwidth     | grows with probe diameter                                                                      | grows with wire size                                                        |
| Power         | door-knob probe: more area, more power                                                         | grows with loop diameter                                                    |
| Removal       | same probe, reverse action                                                                     | same loop, induced current                                                  |



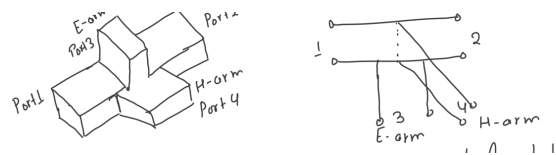
Probe (top) launches E lines; loop (bottom) links H lines.  
Source: Gangaju, slides 3 and 6.

- **Slot / aperture:** a hole in the wall where the field is strong; gives **loose** coupling, used between a guide and a cavity or a second guide (directional coupler, §4.8).
- Choice depends on: voltage breakdown near an antinode, ease of adjustment, constancy of coupling under mechanical change, not disturbing an electron beam, matching.

4.5 Waveguide Tee Junctions and the Magic Tee

Explain the magic tee with a neat diagram and derive its S-matrix; E-plane vs H-plane tee 3+3 4 5 6 8

TOP 9/24 71 Ma, 71 Bh, 72 Ash, 72 Ma, 79 Ch, 79 Bh, 80 Ch, 80 Bh, 82 Bh · construction, derivation and use; identification is §4.6, the radar problems §4.7



Magic tee: collinear ports 1, 2; E-arm 3; H-arm 4.

Construction

- A **hybrid** of the two simple tees on one junction:
  - **E-plane (series) tee:** side arm on the **broad** wall, its axis parallel to the main guide's E field. Port 3.
  - **H-plane (shunt) tee:** side arm on the **narrow** wall, in the plane of H. Port 4.
- Collinear arms 1 and 2 are symmetric about both side arms.
- A **post and iris** inside the junction match ports 3 and 4; without them the junction is simply a “hybrid tee” and is poorly matched.
- It is called **magic** because a matched 4-port with this symmetry turns out to be matched at **all** ports and isolates the two pairs, which nothing in its shape makes obvious.

**Derivation of the S-matrix, the 4-mark version asked in 80 Bh and 72 Ma**

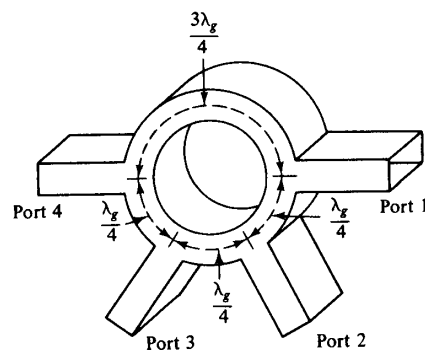
- 4 ports  $\Rightarrow$  a  $4 \times 4$  matrix  $[S]$ ; write all sixteen  $S_{ij}$  out,  $i, j$  from 1 to 4.
- H-arm symmetry**: input at 4 reaches 1 and 2 equally and **in phase**  $\Rightarrow S_{14} = S_{24}$ .
- E-arm antisymmetry**: input at 3 reaches 1 and 2 equally and in **antiphase**  $\Rightarrow S_{13} = -S_{23}$ .
- E and H arms are orthogonal**: an E-arm wave cannot excite the H-arm mode  $\Rightarrow S_{34} = S_{43} = 0$ .
- Matched side arms** (post and iris)  $\Rightarrow S_{33} = S_{44} = 0$ .
- Reciprocity**  $\Rightarrow S_{ij} = S_{ji}$ .
- Unitary, column 3**:  $|S_{13}|^2 + |S_{23}|^2 = 1$ , with  $|S_{13}| = |S_{23}| \Rightarrow S_{13} = 1/\sqrt{2}$ . Same for column 4:  $S_{14} = 1/\sqrt{2}$ .
- Unitary, column 1**:  $|S_{11}|^2 + |S_{12}|^2 + \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow S_{11} = S_{12} = 0$ ; column 2 gives  $S_{22} = 0$ .
- Result:  $[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ , matched at all four ports, lossless, reciprocal.

**Properties (read straight off the matrix)**

- Into **H-arm** (4): equal halves at 1 and 2, **in phase**; nothing at 3.
- Into **E-arm** (3): equal halves at 1 and 2, **antiphase**; nothing at 4.
- Into **collinear** port 1: halves at 3 and 4; **nothing at 2**.
- Equal in-phase signals at 1 and 2: **sum** emerges at 4, nothing at 3.
- Equal antiphase signals at 1 and 2: **difference** emerges at 3, nothing at 4.
- All four ports matched: no reflection anywhere.

**Applications**

- Power combiner, e.g. two radar transmitters into one antenna (§4.7).
- Duplexer: transmitter and receiver on the isolated E and H arms (below).
- Balanced mixer: signal on one side arm, local oscillator on the other.
- Impedance bridge / measurement: unknown and standard on 1 and 2, null detector on 3.
- Monopulse radar comparator: sum and difference channels.

**Hybrid ring (rat race)**

- Annular line, circumference  $1.5\lambda_g$ , four arms at  $\lambda_g/4$ ,  $\lambda_g/4$ ,  $\lambda_g/4$  and  $3\lambda_g/4$ .
- Into port 1: the two ways round to port 3 differ by  $\lambda_g/2$  ( $180^\circ$ )  $\rightarrow$  cancel; to 2 and 4 they add. Same behaviour as a magic tee, in planar form, but only at the design frequency.

**E-plane tee vs H-plane tee**

|                  | <b>E-plane (series)</b>        | <b>H-plane (shunt)</b>     |
|------------------|--------------------------------|----------------------------|
| Side arm on      | broad wall, $\parallel$ E      | narrow wall, $\parallel$ H |
| Side-arm input   | antiphase at 1, 2              | in phase at 1, 2           |
| Relation         | $S_{13} = -S_{23}$             | $S_{13} = S_{23}$          |
| Equivalent       | series junction                | shunt junction             |
| In-phase at 1, 2 | cancel in side arm             | add in side arm            |
| Matrix           | $-1/\sqrt{2}$ in row 3 (below) | $-1/2$ at $S_{12}$ (below) |

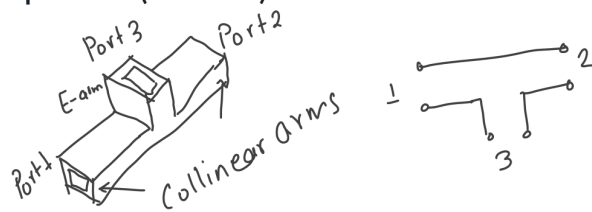
- Both are 3-ports and **cannot** be lossless, reciprocal and matched at once (§3.5): ports 1 and 2 stay mismatched with  $|S_{11}| = 1/2$ .

69 Bh 5, 80 Ba 4

## Prepare the S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses 4 8

PIN 2/24 80 Bh, 81 Ba · derived line by line in the companion, Problems 6 and 7

### E-plane tee (series tee)



- Side arm in the plane of the **E** field, so it excites the collinear arms in **antiphase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the  $-1/\sqrt{2}$  in the third row / column.
- Use:** power splitting with a phase reversal; the difference arm of a magic tee.
- Ports 1 and 2 are not matched** ( $S_{11} = S_{22} = 1/2$ ), which is the 3-port theorem of §3.5 showing up in the answer.

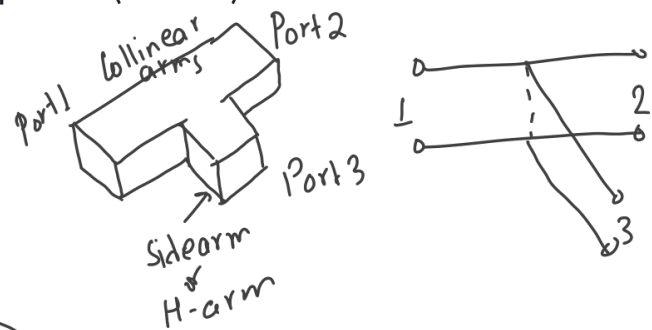
### Duplexer, the third matrix 81 Ba asks for

- Circulator duplexer** (the usual answer): port 1 transmitter, port 2 antenna, port 3 receiver; power circulates  $1 \rightarrow 2 \rightarrow 3$ :

$$[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- Matched ( $S_{ii} = 0$ ), lossless (each column sums to 1) and **non-reciprocal** ( $S_{21} = 1$ ,  $S_{12} = 0$ ): the only way a 3-port can be both matched and lossless (§3.5).
- All** the transmitter power reaches the antenna, **all** the echo reaches the receiver, and  $S_{31} = 0$  keeps the transmitter out of the receiver.

### H-plane tee (shunt tee)



- Side arm in the plane of the **H** field, so the collinear arms come out **in phase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the minus sign has moved to  $S_{12}$ ; the side-arm column is all positive.
- Use:** in-phase power divider or combiner; the sum arm of a magic tee.
- E-tee vs H-tee in one line:** same geometry, same magnitudes, opposite phase relation. The E-tee subtracts, the H-tee adds.

### Magic-tee duplexer (76 Bh, §4.7)

- Transmitter on the H-arm (4), receiver on the E-arm (3), antenna on arm 1, matched dummy load on arm 2. Strike row and column 2; ports (1, 3, 4) = (antenna, Rx, Tx):

$$[S] = \begin{bmatrix} 0 & r & r \\ r & 0 & 0 \\ r & 0 & 0 \end{bmatrix}, \quad r = \frac{1}{\sqrt{2}}$$

- $S_{34} = 0$ : transmitter and receiver **isolated**. But only **half** the transmitter power reaches the antenna, and half the echo goes back to the transmitter: a 6 dB round-trip penalty the circulator avoids.
- Uses:** radar and any single-antenna transceiver; protects the receiver front end from the transmit pulse.

## Tee combinations: arms terminated, two tees joined, a 3-port coupler 4 4+4+2 5 8 10

HOT 5/24 70 Ma, 75 Bh, 77 Ch, 78 Ch, 82 Ba · every matrix below is asserted in wg.py; companion Problems 9 to 11 repeat them as step-by-step eliminations

### Two rules carry every one of these questions

- Start from the magic tee above: ports 1, 2 collinear (the **co-planar** arms), 3 = E-arm, 4 = H-arm,  $r = 1/\sqrt{2}$ ,  $[S_T] = r \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ .
- Terminate** port  $k$  in a load  $\Gamma_k$ :  $a_k = \Gamma_k b_k$ . A **matched** load has  $\Gamma_k = 0$ , so  $a_k = 0$ : column  $k$  is multiplied by zero, and  $b_k$  is absorbed, never seen outside  $\Rightarrow$  **strike out row  $k$  and column  $k$** ; nothing else changes.
- Join** port  $p$  of one junction to port  $q$  of another:  $a_q = b_p$  and  $a_p = b_q$ , what leaves one enters the other. Drive one outside port at a time with  $a = 1$ , follow the wave through, read off one column.

**A. Hybrid tee with a terminated arm** 82 Ba 4+4+2, 70 Ma 8

**(a) E-arm terminated:** strike row and column 3; ports (1, 2, 4)

$$[S_a] = \begin{bmatrix} 0 & 0 & r \\ 0 & 0 & r \\ r & r & 0 \end{bmatrix}$$

- **Matched:**  $S_{11} = S_{22} = S_{44} = 0$ . **Reciprocal:**  $[S_a] = [S_a]^T$ .
- **H-arm in:**  $b_1 = b_2 = r$ , equal (−3 dB each) and **in phase**; column 4 power  $\frac{1}{2} + \frac{1}{2} = 1$ , no loss  $\Rightarrow$  an **in-phase 3 dB power divider**.
- **Collinear isolation kept:**  $S_{12} = 0$ .
- **Port 1 in:** only  $|S_{41}|^2 = \frac{1}{2}$  reaches port 4; the other half ( $S_{31} = r$ ) runs into the E-arm load. Column 1 power  $\frac{1}{2} \Rightarrow$  **not lossless**.
- **As a combiner:** equal in-phase inputs at 1 and 2 add fully at 4; any antiphase (difference) part is dumped in the load.
- **Why (a) and (b) are lossy:** each is a matched, reciprocal 3-port, and §3.5 proves no such 3-port can be lossless. The load is where the missing half goes, and only a drive from a collinear arm reaches it.

**(c) Both co-planar arms terminated:** strike rows and columns 1, 2; ports (3, 4)

$$[S_c] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

- $S_{33} = S_{44} = 0$ : E-arm and H-arm both **matched**.
- $S_{34} = 0$ : E and H **completely isolated**.
- Power into either side arm splits between the two loads (antiphase from E, in phase from H) and none comes out: the tee is a **matched termination** for both arms.

**(b) H-arm terminated:** strike row and column 4; ports (1, 2, 3)

$$[S_b] = \begin{bmatrix} 0 & 0 & r \\ 0 & 0 & -r \\ r & -r & 0 \end{bmatrix}$$

- Matched and reciprocal, as in (a).
- **E-arm in:**  $b_1 = r$ ,  $b_2 = -r$ , equal and 180° **apart**; column 3 power 1  $\Rightarrow$  an **antiphase 3 dB divider**, a balanced feed.
- Ports 1 and 2 isolated,  $S_{12} = 0$ .
- **Port 1 in:** half to the E-arm, half ( $S_{41} = r$ ) lost in the H-arm load; column 1 power  $\frac{1}{2}$ .
- **As a combiner:** an antiphase pair at 1, 2 leaves entirely by 3; any in-phase (sum) part is absorbed.

**Worth writing for the 2 marks:** the isolation does not need matched loads.

- Loads  $\Gamma_1, \Gamma_2$  on arms 1, 2. E in ( $a_3 = 1$ ):  $b_1 = r$ ,  $b_2 = -r$ ; reflected  $a_1 = r\Gamma_1$ ,  $a_2 = -r\Gamma_2$ ; then  $b_3 = ra_1 - ra_2$ ,  $b_4 = ra_1 + ra_2$ :

$$S_{33} = S_{44} = \frac{\Gamma_1 + \Gamma_2}{2}, \quad S_{34} = \frac{\Gamma_1 - \Gamma_2}{2}$$

- $\Gamma_1 = \Gamma_2 \Rightarrow S_{34} = 0$  whatever the loads: E and H **stay isolated**. This is the **impedance bridge**: unknown on arm 1, standard on arm 2, source on E, detector on H; a null means the two loads are equal.
- $\Gamma_1 = \Gamma_2 = 0$  gives  $[S_c]$ ;  $\Gamma_1 = \Gamma_2 = -1$  (shorts) gives  $-[I]$ , the mirror of the 82 Bh band below.

**B. Two magic tees joined** 77 Ch 10 E-arm to E-arm, 78 Ch 5 H-arm to H-arm

- Tee A ports 1, 2, 3, 4; tee B ports 1', 2', 3', 4'. **77 Ch** joins the E-arms:  $a_{3'} = b_3$ ,  $a_3 = b_{3'}$ . Ports 3 and 3' vanish; six remain, 1, 2, 4, 1', 2', 4'  $\Rightarrow$  a  $6 \times 6$  matrix.
- **Drive port 1** ( $a_1 = 1$ ): tee A gives  $b_2 = 0$ ,  $b_4 = r$ ,  $b_3 = r$ . That  $r$  enters tee B's E-arm:  $b_{1'} = S_{13}r = \frac{1}{2}$ ,  $b_{2'} = S_{23}r = -\frac{1}{2}$ ,  $b_{4'} = S_{43}r = 0$ . Nothing comes back, since tee B's E-arm is matched. Column 1 =  $(0, 0, r, \frac{1}{2}, -\frac{1}{2}, 0)$ , power  $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$  ✓
- **Port 2:**  $b_3 = S_{32} = -r \Rightarrow b_{1'} = -\frac{1}{2}$ ,  $b_{2'} = +\frac{1}{2}$ ; and  $b_4 = r$ .
- **Port 4 (H-arm A):**  $b_1 = b_2 = r$  but  $b_3 = S_{34} = 0 \Rightarrow$  **nothing crosses**.
- Ports 1', 2', 4': mirror images, by the symmetry of the join.

## 77 Ch: E-arm joined to E-arm

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 1' & 2' & 4' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \\ 1' \\ 2' \\ 4' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & r & -\frac{1}{2} & \frac{1}{2} & 0 \\ r & r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & r \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ 0 & 0 & 0 & r & r & 0 \end{bmatrix} \end{matrix}$$

**78 Ch: H-arm joined to H-arm**,  $a_{4'} = b_4$ ,  $a_4 = b_{4'}$ ; outside ports 1, 2, 3, 1', 2', 3'. Port 1:  $b_4 = r$  enters tee B's H-arm  $\Rightarrow b_{1'} = b_{2'} = +\frac{1}{2}$ , in phase; port 2 gives the same.

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 1' & 2' & 3' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 1' \\ 2' \\ 3' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -r & \frac{1}{2} & \frac{1}{2} & 0 \\ r & -r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -r \\ 0 & 0 & 0 & r & -r & 0 \end{bmatrix} \end{matrix}$$

## C. 3-port directional coupler 75 Bh 4

- Take the ideal 4-port coupler of §3.6: port 1 input, 2 through, 3 coupled, 4 isolated,  $\alpha^2 + \beta^2 = 1$ .
- A **3-port coupler** is that coupler with port 4 terminated in a matched load **inside the casing**. Strike row and column 4:

$$[S] = \begin{bmatrix} 0 & \alpha & j\beta \\ \alpha & 0 & 0 \\ j\beta & 0 & 0 \end{bmatrix}$$

- Coupling  $C = -20 \log \beta$  dB; insertion loss  $-20 \log \alpha$  dB.
- Use: forward-power monitor, where only the forward sample is wanted.

## verified/added Magic tee with both E- and H-arms short-circuited 4

PIN 1/24 82 Bh · not in any source note, derived here and checked in `sparam.py`; companion Problem 12

- Shorts at the junction reference plane,  $\Gamma = -1$ :  
 $a_3 = -b_3$ ,  $a_4 = -b_4$ .
- Magic-tee rows 3 and 4:  $b_3 = r(a_1 - a_2)$ ,  
 $b_4 = r(a_1 + a_2)$ .
- Rows 1 and 2 with the shorts substituted:

$$b_1 = ra_3 + ra_4 = -r(b_3 + b_4) = -r^2(2a_1) = -a_1$$

$$b_2 = -ra_3 + ra_4 = r(b_3 - b_4) = r^2(-2a_2) = -a_2$$

- So the 4-port collapses to the two-port seen from ports 1 and 2:

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

## Properties (77 Ch)

- Matched**: all six  $S_{ii} = 0$ . **Reciprocal**:  $[S] = [S]^T$ .
- Lossless**: every column's powers sum to 1; nothing is loaded.
- Collinear isolation kept:  $S_{12} = S_{1'2'} = 0$ . The two H-arms are isolated,  $S_{44'} = 0$ .
- Port 1 to port 1':  $|S|^2 = \frac{1}{4}$ ,  $-6$  dB, two 3 dB splits in cascade.
- Difference crosses**: an antiphase pair  $(r, -r)$  at 1, 2 reappears whole as  $(r, -r)$  at 1', 2', nothing at 4.
- Sum stays**: an in-phase pair  $(r, r)$  at 1, 2 leaves entirely by its own H-arm 4; nothing crosses.
- Use: a sum/difference separator that hands one tee's difference channel to a second tee.

## E-join vs H-join, the short-note table

|               | E to E (77 Ch)                                 | H to H (78 Ch)            |
|---------------|------------------------------------------------|---------------------------|
| Crosses over  | difference (antiphase)                         | sum (in phase)            |
| At 1', 2'     | in antiphase                                   | in phase                  |
| Stays home    | sum, via own H-arm                             | difference, via own E-arm |
| Isolated pair | the two H-arms                                 | the two E-arms            |
| Both          | matched at all six ports, reciprocal, lossless |                           |

- Matched** ( $S_{ii} = 0$ ) and **reciprocal**.
- Forward** (port 1 in):  $\alpha^2$  to 2,  $\beta^2$  to 3; column power 1, **no loss**.
- Ports 2 and 3 isolated**:  $S_{23} = 0$ .
- Backward** (port 2 in): only  $\alpha^2$  returns to 1; the coupled share  $\beta^2$  goes to the internal load. Port 3 in:  $\beta^2$  to 1,  $\alpha^2$  absorbed.
- Must be lossy**: matched and reciprocal 3-port  $\Rightarrow$  not lossless (§3.5); the loss shows only for backward drive.
- 10 dB coupler:  $\beta = 0.316$ ,  $\alpha = 0.949$ . Forward: 90% through, 10% coupled. Backward into 2: 90% reaches 1, 10% absorbed.

- Behaviour**: each collinear arm reflects **all** of its incident power with a  $180^\circ$  phase reversal, and the two arms stay **completely isolated** from each other.
- The E-arm and H-arm reflections add in the driven arm and cancel in the other one, which is why  $S_{21}$  stays zero.
- Still lossless (each column sums to 1), still reciprocal. The tee has become a **pair of independent short circuits**, useful as a reflective phase shifter or a reference short.

4.6 Identifying a Device from its S-Matrix

Identify and explain, with a neat diagram, the passive microwave device described by the given S-matrix 5 6 8 10

HOT 4/24 74 Bh, 75 Bh, 78 Ch, 81 Bh · in all four papers the printed matrix is a magic tee

The six-step method

1. **Count the ports:** the order of the matrix is the port count.  $4 \times 4$  narrows it to coupler, magic tee or hybrid ring immediately.
2. **Read the diagonal:** zeros mark the matched ports. An all-zero diagonal on a 4-port means matched everywhere.
3. **Test the transpose:** symmetric means passive and isotropic; asymmetric means ferrite or active.
4. **Test the columns:** sums of 1 and orthogonal pairs mean lossless. A shortfall is dissipation.
5. **Map the zeros:** an off-diagonal zero is an **isolated pair**. Which pairs are decoupled is the strongest single clue.
6. **Read the signs and the  $j$ 's:** a sign flip down a column means an **antiphase (difference, E) arm**; matching signs mean an **in-phase (sum, H) arm**; a factor  $j$  means a **quadrature** coupled arm.

Answer skeleton

- Name the device in the first line. Do not narrate the tests first.
- Draw it: the 3D waveguide sketch with all arms labelled, plus the schematic.
- Justify with the four tests, one line each.
- Close with two uses and one limitation (the tee needs matching irises at the collinear arms to reach the ideal matrix in practice).

Fingerprint table

| Pattern                                                                                      | Device                                         |
|----------------------------------------------------------------------------------------------|------------------------------------------------|
| $2 \times 2$ ,<br>$S_{11} = S_{22} = 0$ ,<br>$S_{12} = S_{21} = 1$                           | Matched lossless line or ideal matched network |
| $2 \times 2$ , $S_{12} = 0$ ,<br>$S_{21} = 1$                                                | Isolator                                       |
| $2 \times 2$ , $ S_{21}  > 1$ ,<br>$ S_{12} $ tiny                                           | Amplifier (active, not reciprocal)             |
| $3 \times 3$ , cyclic ones,<br>not symmetric                                                 | Circulator                                     |
| $3 \times 3$ with a $\pm 1/\sqrt{2}$<br>column                                               | E-plane tee (–) or H-plane tee (+)             |
| $4 \times 4$ , zero diagonal,<br>two zero $2 \times 2$<br>blocks, one column<br>sign-flipped | <b>Magic tee</b>                               |
| $4 \times 4$ , zero diagonal,<br>$j$ on the coupled<br>arm, one zero per<br>row              | Directional coupler                            |

### The three matrices the four papers print

- 78 Ch 10, printed without its normalising factor:

$$[S] = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

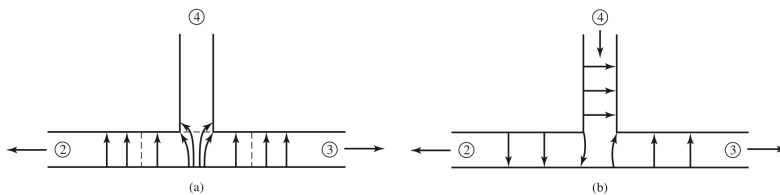
Zero diagonal, symmetric, the (1, 2) and (3, 4) blocks zero, column 3 sign-flipped: **magic tee**. It is unitary only after dividing by  $\sqrt{2}$ , and saying so earns the “judge the condition” mark.

- 81 Bh 5 and 74 Bh 8, printed symbolically:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

Column 3 in phase (H-arm), column 4 antiphase (E-arm),  $S_{33} = S_{44} = 0$ : the **magic tee before the unitary property is applied**. Finish the job and  $S_{11} = S_{22} = S_{12} = 0, S_{13} = S_{14} = 1/\sqrt{2}$ .

### Why the signs fall where they do



- 75 Bh 6, the same device with the collinear isolation already written in:

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$$

Here column 3 is the antiphase E-arm and column 4 the in-phase H-arm, the mirror of the previous labelling.

**Read the signs, not the port numbers.**

- Not to be confused with 82 Bh 4, which prints two *amplifier* sets: steps 3 and 4 answer it ( $S_{12} \neq S_{21}$ ,  $|S_{21}| > 1$ ), and it is a Chapter 3 question, §3.4 and Ch3 companion Problem 2.

*E* in the collinear arms (2), (3) of a magic tee. (a) fed from the H-arm (Pozar's port 1, the sum port): the arms are driven **in phase**, the matching-sign column. (b) fed from the E-arm (4): **antiphase**, the sign-flipped column. Pozar numbers the arms differently from the papers, so label by E and H, not by number. Pozar Fig 7.50.

## 4.7 Power Splitting and Combining: the Radar Questions

Two identical radar transmitters must deliver twice the power to one antenna; and an ASR needs half the transmitter power at the antenna through a duplexer. Solve both with an S-matrix 6 8 10

**HOT 4/24** 73 Ma, 73 Bh, 74 Ma, 76 Bh · one device, two directions: the magic tee combines and splits

### Combining: two transmitters into one antenna

- **Device:** magic tee (or a 3 dB hybrid coupler).
- **Connection:** the two transmitters drive the collinear arms 1 and 2 **in phase and at equal amplitude**; the antenna hangs on the H-arm (port 4); the E-arm (port 3) carries a **matched dummy load**.
- **S-matrix** of the magic tee (§4.5; collinear 1, 2, E-arm 3, H-arm 4):

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- **Drive vector:** equal amplitude, in phase, so  $a_1 = a_2 = a$  and  $a_3 = a_4 = 0$ .
- **Multiply out,**  $[b] = [S][a]$ :

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ a \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \sqrt{2} a \end{bmatrix}$$

### Combining: reading the answer off $[b]$

- **Read each row:**  $b_4 = \frac{1}{\sqrt{2}}(a_1 + a_2)$ , the sum at the H-arm;  
 $b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$ , the difference at the E-arm;  
 $b_1 = b_2 = 0$  because  $S_{11} = S_{22} = S_{12} = 0$ .
- **Powers,** with  $P = |a|^2$  per transmitter:

$$|b_4|^2 = 2|a|^2 = 2P, \quad |b_1|^2 = |b_2|^2 = |b_3|^2 = 0$$

- **Result:** the antenna receives  $2P$ , twice what either transmitter can deliver; the dummy load receives nothing, and nothing reflects back into either transmitter. Power in  $2P$  = power out  $2P$ : lossless.
- **Why the tee and not a plain T-junction:**  $S_{12} = 0$ , so the two transmitters are **isolated** and cannot pull or damage each other; and all four ports stay matched, so neither transmitter sees a VSWR.
- **Practical riders** worth a mark each:
  - Phase must be equalised, so the two feed lines are cut to the same electrical length.
  - Amplitude imbalance  $\Delta$  shows up as  $|b_3|^2 \neq 0$ : the dummy load absorbs exactly the mismatch, which is what protects the transmitters.
  - The dummy load must be rated for the full unbalanced power.

**Splitting: half the transmitter power to the antenna**

- **Device:** the same magic tee, used as a duplexer, driven from the H-arm.
- **Connection:** transmitter on port 4 (H-arm), antenna on port 1, matched load on port 2, receiver on port 3 (E-arm).
- **S-matrix of the duplexer**, which is what **76 Bh** asks you to prepare: the magic tee's, with rows and columns in the order (1 Ant, 2 Load, 3 Rx, 4 Tx):

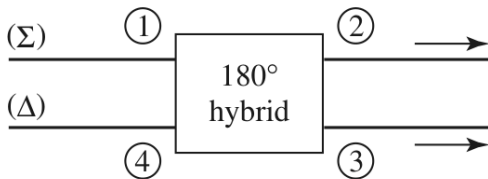
$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- **Transmit**,  $a_4 = a$  and all other inputs zero:

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ a \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} a \\ a \\ 0 \\ 0 \end{bmatrix}$$

$$|b_1|^2 = |b_2|^2 = \frac{1}{2}|a|^2, \quad |b_3|^2 = |b_4|^2 = 0$$

- **Result:** the antenna gets exactly  $P/2$ , as **76 Bh** requires, the load takes the other  $P/2$ , and the receiver gets **none** of the transmitter power ( $S_{34} = 0$ ), which is the whole point of a duplexer.

**The schematic to draw beside the tee****Splitting: the receive path**

- **Receive**, echo  $a_1 = e$  and all other inputs zero:

$$[b] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ e \\ e \end{bmatrix} \Rightarrow |b_3|^2 = \frac{1}{2}|e|^2 \text{ to the receiver}$$

The other half goes back to the transmitter. The 3 dB penalty in each direction is the price of the magic-tee duplexer, and naming it shows you understand the device rather than reciting it.

- **Better answer for a real ASR: a ferrite circulator** duplexer routes the full power  $1 \rightarrow 2$  and the full echo  $2 \rightarrow 3$  with no 3 dB loss. Quote it as the improvement whenever the question does not force the half-power split.

**Answer skeleton for both**

- Name the device, draw it with every arm labelled and every termination named.
- Write  $[S]$  for the magic tee.
- Write the drive vector  $[a]$  the connection creates.
- Multiply out, and read the answer off  $|b|^2$ .
- Close with the isolation and matching argument: that is the "sufficient explanation" the marks are for.

- A magic tee is a  $180^\circ$  hybrid: **sum** port  $\Sigma$  (the H-arm) and **difference** port  $\Delta$  (the E-arm), with outputs (2) and (3).
- **Combining:** equal in-phase inputs at (2), (3) add at  $\Sigma$  and cancel at  $\Delta$ , so the dummy load on  $\Delta$  sees nothing.
- **Splitting:** drive  $\Sigma$  and (2), (3) each get half, in phase;  $\Delta$  stays isolated.
- Pozar Fig 7.41.

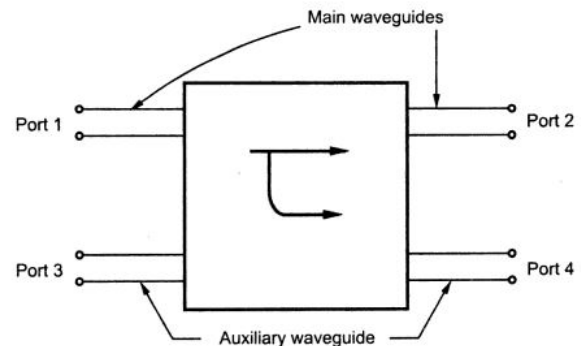
## 4.8 Directional Couplers

Explain directional couplers and the four parameters used to describe their performance 4 5 6

HOT 3/24 73 Bh, 79 Ch, 80 Ba

### What it is

- A **4-port** junction that samples power travelling in **one direction** only.
- Main guide: port 1 (input) to port 2 (through).
- Auxiliary guide: port 4 (coupled) and port 3 (isolated), Gangaju's numbering.
- Ideal, all ports matched:
  - $1 \rightarrow 2$  passes almost all the power.
  - $1 \rightarrow 4$  a known small fraction.
  - $1 \rightarrow 3$  nothing.
- Uses: **power monitoring**, **reflectometers** and **VSWR** measurement (one coupler each way), network analysers, signal injection, and as a 3 dB hybrid.
- S-matrix: §3.6. Note §3.6 labels the coupled port 3; the numbering varies by text, the physics does not.



Block view, Gangaju slide 54.

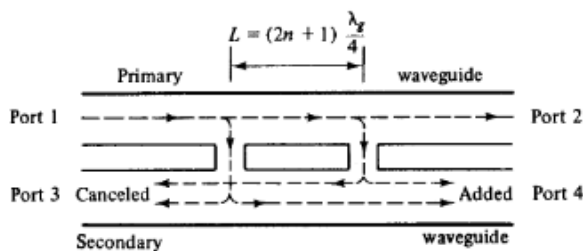
The four parameters ( $P_i$  = power at port  $i$ , input at 1)

| Parameter          | Definition                   | Measures                           |
|--------------------|------------------------------|------------------------------------|
| Coupling<br>$C$    | $10 \log \frac{P_1}{P_4}$ dB | how much is sampled                |
| Directivity<br>$D$ | $10 \log \frac{P_4}{P_3}$ dB | forward vs backward discrimination |
| Isolation $I$      | $10 \log \frac{P_1}{P_3}$ dB | leakage to isolated port           |
| Insertion loss     | $10 \log \frac{P_1}{P_2}$ dB | loss on the main line              |

- $I = C + D$ . Example:  $P_1 = 10$  mW,  $P_4 = 1$  mW,  $P_3 = 0.1$   $\mu$ W  $\Rightarrow C = 10$  dB,  $D = 40$  dB,  $I = 50$  dB.
- Good coupler:  $D$  of 30 to 40 dB, VSWR near 1.
- **verified/added Frequency sensitivity** (coupling flatness): how far  $C$  drifts over the specified band. A fifth figure some notes list; quote it after the four.

### Two-hole coupler: why it is directional

- Two small holes in the common wall, spaced  $d = (2n + 1)\lambda_g/4$ .
- Each hole radiates a small wave into the auxiliary guide, both ways.
- **Forward** waves (towards port 4) travel equal total paths  $\rightarrow$  **in phase**  $\rightarrow$  add.
- **Backward** waves (towards port 3) differ by  $2d = (2n + 1)\lambda_g/2 \rightarrow (2n + 1)\pi$  out of phase  $\rightarrow$  **cancel**.
- Exact only at one frequency; more holes with tapered sizes widen the band.



Two-hole coupler, Gangaju slide 57.

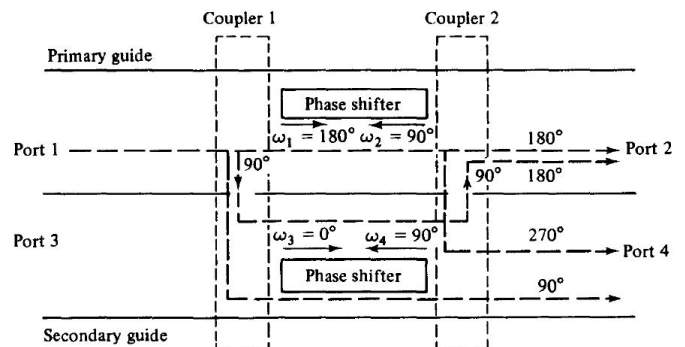
## 4.9 Ferrite Devices: Circulators, Isolators, Phase Shifters

### Explain the microwave circulator 5

HOT 3/24 70 Bh, 72 Ash, 73 Ma · asked word for word each time

#### Circulator

- A **non-reciprocal** multi-port in which power entering port  $n$  leaves only by port  $n + 1$ :  $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$  (3-port), or round four ports.
- Ideal: **matched** at every port and **lossless**, but  $S_{ij} \neq S_{ji}$ .  
3-port form (matrix of §3.5):  $[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ .
- Non-reciprocity comes from a **magnetised ferrite**.
- **Faraday rotation**: a plane wave along a ferrite rod in an axial magnetic field has its polarisation rotated by an angle set by the field strength and the rod length, and in the **same absolute sense** whichever way it travels. A return trip therefore doubles the rotation instead of undoing it.
- **Four-port construction** (slide 62): two 3 dB side-hole couplers, a primary and a secondary guide, and a non-reciprocal ferrite phase shifter in each.
  - Port 1  $\rightarrow$  port 2: both paths arrive with the same phase  $\rightarrow$  add.
  - Port 1  $\rightarrow$  port 4: the paths differ by  $180^\circ \rightarrow$  cancel.
  - The phase shifters behave differently in the reverse direction, so port 2 feeds port 3, port 3 feeds 4, port 4 feeds 1.



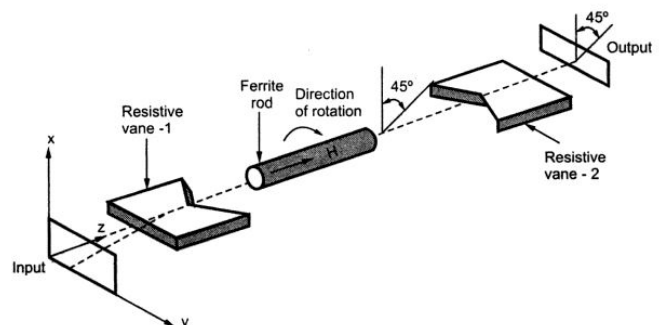
Four-port circulator: two 3 dB couplers plus ferrite phase shifters. Gangaju slide 62.

#### Applications

- **Duplexer**: transmitter port 1, antenna 2, receiver 3 (§4.5).
- Reflection amplifiers (tunnel diode, parametric): separate input from output.
- An **isolator**, by loading one port.

#### Isolator (uniline)

- Passes power one way, **absorbs** it the other way:  $[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ .
- Circulator with port 3 matched-terminated, or a Faraday-rotation isolator:
  - Input E vertical; ferrite rotates it  $45^\circ$ ; output guide twisted  $45^\circ$  to accept it.
  - Reflection from the load is rotated a further  $45^\circ$  in the same sense  $\rightarrow 90^\circ$  from the input orientation  $\rightarrow$  absorbed by a resistive vane.
- Use: between a klystron or magnetron and its load, so a mismatch cannot reflect back and pull the oscillator frequency.



### Phase shifter (syllabus, not yet asked)

- Changes the phase of the transmitted wave by  $\Delta\phi = 2\pi l/\lambda_g$ , either by changing the **length**  $l$  or the **guide wavelength**  $\lambda_g$ .
- Mechanical: telescoping coaxial line; **dielectric vane** moved across the guide (more dielectric in the high-E centre, smaller  $\lambda_g$ , more phase).
- Electrical: **ferrite** under a variable bias field; PIN-diode switched lines.
- Uses: phased-array beam steering, phase discriminators, power-amplifier linearisation.

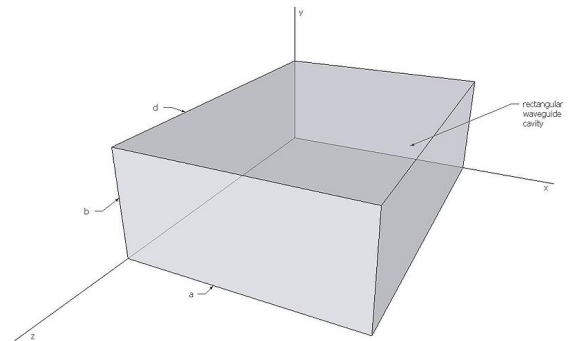
## 4.10 Cavity Resonators

### Explain microwave cavity resonators 4 6

PIN 2/24 74 Ma, 74 Bh

#### What and why

- A closed metal enclosure that stores EM energy at discrete resonant frequencies: a section of waveguide shorted at **both** ends.
- Microwave replacement for the LC tank: lumped L and C fail above a few hundred MHz (radiation, skin effect, lead parasitics).
- Circuit analogue: stored E energy  $\rightarrow$  C, stored H energy  $\rightarrow$  L, wall loss  $\rightarrow$  R.
- Types: **rectangular**, **circular**, **re-entrant** (the klystron cavity, Ch 5).
- Infinitely many modes; the lowest is the **dominant** mode.



Rectangular cavity  $a \times b \times d$ . Gangaju slide 90.

#### Rectangular cavity: resonant frequency

- Standing wave along  $z$ : the guide length must hold  $p$  half guide-wavelengths,  $d = p\lambda_g/2 \Rightarrow \beta = p\pi/d$ .
- $k^2 = k_c^2 + \beta^2$  gives

$$f_{mnp} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

- Allowed indices:
  - TE<sub>mnp</sub>:  $p \geq 1$ ,  $m$  and  $n$  not both zero.
  - TM<sub>mnp</sub>:  $m, n \geq 1$ ,  $p \geq 0$ .
- Dominant for  $b < a < d$ : **TE<sub>101</sub>**,  $f_{101} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}}$ .  
Example  $2 \times 1 \times 3$  cm: 9.01 GHz, next TE<sub>102</sub> at 12.5 GHz.
- Circular cavity, radius  $a$ , length  $d$ :

$$f = \frac{c}{2\pi} \sqrt{\left(\frac{p'_{nm}}{a}\right)^2 + \left(\frac{l\pi}{d}\right)^2} \text{ (TE), with } p_{nm} \text{ for TM.}$$

#### Quality factor

- $Q = \omega \frac{\text{energy stored}}{\text{power lost}}$ .
- Cavities reach  $10^4$  to  $10^6$ ; a lumped LC circuit about  $10^2$ .
- Losses add as reciprocals:

$$\frac{1}{Q_L} = \frac{1}{Q_c} + \frac{1}{Q_d} + \frac{1}{Q_{ext}}$$

- $Q_c$ : wall conduction loss.
- $Q_d = 1/\tan\delta$ : dielectric filling.
- $Q_{ext}$ : power taken by the coupling;  $Q_L$  is the loaded Q.
- Coupling (probe, loop, aperture) is **under-, critically or over-coupled**; at critical coupling the input is matched.

#### Applications

- Klystron and magnetron cavities; wavemeters; filters; oscillator stabilisation; TR cells in duplexers.

#### verified/added Tuning a cavity

- Change the stored field and  $f_0$  moves: a **plunger** alters the length  $d$ ; a **post or screw** inserted where E is strongest adds capacitance and lowers  $f_0$  (**capacitive** tuning); one where H is strongest reduces the inductance and **raises**  $f_0$  (**inductive** tuning).
- Used to set the resonant frequency, match the cavity to its line and trim  $Q$ .

## 4.11 Solid-State and Quantum Devices: Gunn Diode, MASER, Microwave Transistors

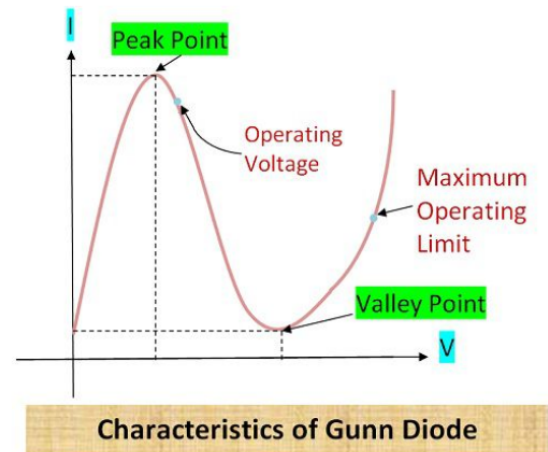
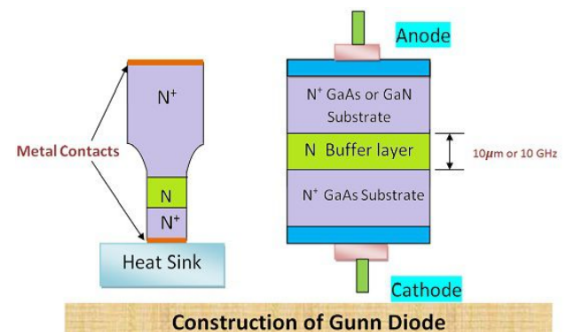
### Gunn diode 4

PIN 1/24 79 Bh · short note

- **Transferred-electron device (TED):** bulk n-type GaAs (or InP), **no p-n junction**; “diode” only because it has two terminals.
- **Construction:**  $n^+$  contact / lightly doped n active layer (epitaxial, about 6 to 18  $\mu\text{m}$ ) /  $n^+$  substrate, metal contacts, heat sink.
- **Two-valley model** (Ridley-Watkins-Hilsum):
  - GaAs conduction band has a **lower valley** (light electrons, high mobility) and an **upper satellite valley** about 0.36 eV higher (heavy, low mobility).
  - Low field: electrons stay in the lower valley, current rises with voltage.
  - Above a threshold field (about 3.2 kV/cm) electrons gain enough energy to **transfer** to the upper valley  $\rightarrow$  average mobility falls  $\rightarrow$  drift velocity **falls as field rises**.
  - That is **negative differential resistance** between the peak and the valley of the I-V curve.
- **Oscillation:** a high-field **domain** forms at the cathode, drifts to the anode at  $v_d \approx 10^5$  m/s, and a new one forms when it arrives:

$$f = \frac{v_d}{L} \quad \text{e.g.} \quad \frac{10^5}{10 \mu\text{m}} = 10 \text{ GHz}$$

- **Correction to the deck** (slide 72): it says the low-field current rises because electrons move from the valence band to the conduction band. It does not; the carriers are already conduction electrons in the lower valley. verified/added



Construction and the peak-valley characteristic. Gangaju slides 69 and 73.

#### (+) and (-)

- (+) small, cheap, low noise, wide band, reliable, 1 to 100+ GHz.
- (-) low efficiency (a few %), poor temperature stability, high bias current.

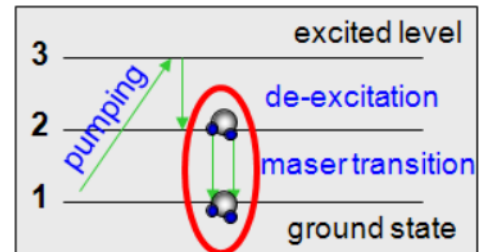
#### Applications

- Local oscillators, police and door-opening radar sensors, low-power transmitters, microwave relay links.

## MASER 4

PIN 1/24 80 Ba · short note

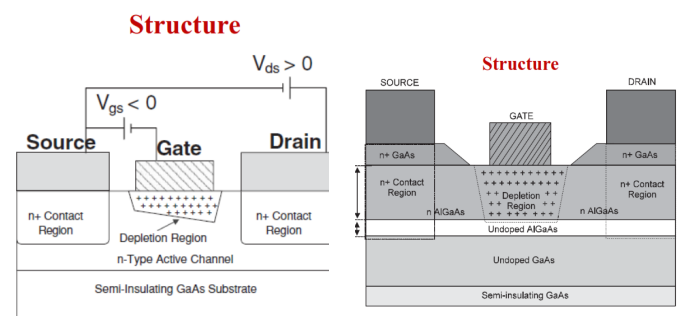
- **Microwave Amplification by Stimulated Emission of Radiation:** the microwave ancestor of the laser (Townes, 1954).
- A **quantum** device working on energy levels of atoms or molecules. The deck's "semiconductor device" is wrong. verified/added
- **Three-level principle:**
  - **Pump** at  $f_{13}$  lifts particles from level 1 to 3.
  - They decay without radiating to the **metastable** level 2 and accumulate → **population inversion** between 2 and 1.
  - An incoming signal at  $f_{21} = (E_2 - E_1)/h$  **stimulates** emission of identical photons (same frequency, phase, direction) → coherent **amplification**.
- **Types:** ammonia gas maser (23.87 GHz, the first), hydrogen maser (1.42 GHz, frequency standard), solid-state ruby maser ( $\text{Cr}^{3+}$  in  $\text{Al}_2\text{O}_3$ , tuned by a magnetic field, cooled to a few kelvin).
- **Key property:** extremely **low noise**.
- **Applications:** radio-telescope and deep-space receivers, satellite ground stations, atomic clocks.



Pump to level 3, relax to 2, stimulated transition 2 to 1.  
Gangaju slide 78.

### Microwave transistors (syllabus, not yet asked)

- Differ from low-frequency parts in **geometry** and **packaging**: interdigitated, overlay and matrix layouts cut lead inductance and junction capacitance.
- **Si BJT**: up to a few GHz. **HBT** (heterojunction, e.g. SiGe, InGaP/GaAs): higher  $f_T$ , amplifiers and MMICs.
- **GaAs MESFET**: Schottky-barrier gate on an n-channel; high electron mobility → well above 5 GHz; low noise, depletion-mode.
- **HEMT**: AlGaAs/GaAs heterojunction confines a 2-D electron gas in undoped GaAs, away from donor scattering → even higher mobility, beyond 20 GHz, lowest noise.



MESFET (left) and HEMT (right) cross-sections. Gangaju slides 85 and 86.

## 4.12 Attenuators, Terminations, Windows, Microstrip

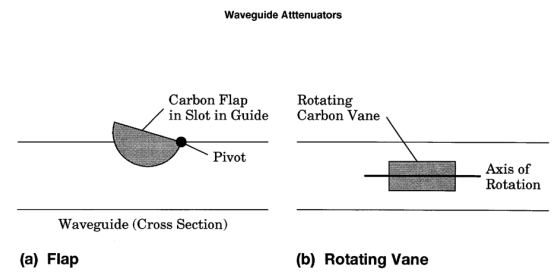
### Microstrips 5

PIN 1/24 69 Bh · geometry and microstrip vs strip-line are in §1.6; stub realisation in §2.9

- Strip of width  $W$  on a substrate of thickness  $h$  and permittivity  $\epsilon_r$ , ground plane underneath, air above.
- **Quasi-TEM**: the field lives partly in air and partly in the substrate, so it sees an effective permittivity  $\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}}$ ,  $1 < \epsilon_{\text{eff}} < \epsilon_r$ .
- $v_p = c/\sqrt{\epsilon_{\text{eff}}}$ ,  $\lambda_g = \lambda_0/\sqrt{\epsilon_{\text{eff}}}$ ;  $Z_0$  falls as  $W/h$  rises.
- (+) planar, cheap photolithography, components mount on top, MMIC friendly.
- (-) radiation at bends, dispersion, lower Q and power than waveguide, crosstalk.

### Attenuators and terminations (syllabus, not yet asked)

- **Attenuator**: reduces power by a set amount with low VSWR; a resistive (carbon) vane or flap in the guide absorbs power through induced currents.
  - **Fixed**: lossy vane, or lossy walls.
  - **Variable**: **flap** lowered through a slot in the centre of the broad wall (deeper = more loss, since  $E$  is strongest there), **vane** moved from side wall to centre, or **rotary** vane in a circular guide.
- **Matched termination**: absorbs all incident power; a guide section ending in a **tapered** absorbing wedge so the impedance changes gradually and nothing reflects.
- Reference **short**: a metal plate; **sliding short**: movable, sets a reactance.



Flap and rotating-vane attenuators. Gangaju slide 39.

### Windows, posts and screws (waveguide matching)

- **Inductive window**: diaphragms from the side walls → shunt  $L$ .
- **Capacitive window**: from top and bottom → shunt  $C$ ; limits power (breakdown).
- **Resonant window**: both → parallel  $LC$ , transparent at resonance.
- **Post / screw**: part-way in → capacitive; right through → inductive;  $\lambda/4$  deep → series resonance.
- Waveguide equivalent of a stub (§2.6).

### 4.13 PYQ Mapping

#### Theory questions, covered in this document

- Rectangular waveguide field equations, TE mode → §4.1. 70 Bh 7, 74 Ma 4, 77 Ch 8, 79 Ch 10, 80 Ch 10, 80 Bh 4
- Rectangular waveguide field equations, TM mode → §4.1. **Most repeated topic of the chapter, with the TE list: 14 papers.** 70 Ma 7, 71 Bh 8, 72 Ma 5, 74 Bh 10, 76 Bh 10, 81 Ba 8, 81 Bh 8
- Rectangular waveguide: modes of propagation and critical parameters → §4.1 Step 7 and §4.2. 69 Bh 8
- Prove  $TM_{01}$  /  $TM_{10}$  do not exist → §4.2. 70 Ma 3
- Lowest TM mode is  $TM_{11}$  → §4.2. 82 Bh 3
- $TE_{10}$  dominant;  $TE_{10}$  vs  $TE_{20}$  → §4.2, with the 81 Bh inequality defect. 71 Bh 2, 81 Bh 3
- Dominant and degenerate modes;  $a = 3b$  → §4.2 and companion Problem 4. 72 Ash 5, 72 Ma 2+2, 82 Ba 3
- Circular waveguide, TE / TM field equations, TE vs TM, dominant mode → §4.3. 71 Ma 10, 75 Bh 8+2, 79 Bh 10, 82 Ba 7
- Probe vs loop coupling → §4.4. 69 Bh 5
- Magic tee: principle, S-parameters, derivation → §4.5 and companion Problem 8. 71 Ma 5, 71 Bh 5, 72 Ash 3+3, 72 Ma 6, 79 Ch 5, 79 Bh 4, 80 Ch 5, 80 Bh 4
- Magic tee with both arms shorted → §4.5 and companion Problem 12. **verified/added** 82 Bh 4
- S-matrices of E-plane tee, H-plane tee and a duplexer, and their uses → §4.5 and companion Problems 6 and 7. 81 Ba 8
- E-plane vs H-plane tee → §4.5. 69 Bh 5, 80 Ba 4
- Hybrid tee; terminated E, H, co-planar arms → §4.5 and companion Problem 11. 70 Ma 8, 82 Ba 4+4+2
- Two magic tees joined at E-arms / in H-plane → §4.5 and companion Problems 9 and 10. 77 Ch 10, 78 Ch 5
- Identify the passive device from a printed S-matrix → §4.6 and companion Problems 13 to 15, all three printed matrices worked. 74 Bh 8, 75 Bh 6, 78 Ch 10, 81 Bh 5
- Radar power combining and duplexer power splitting → §4.7 and companion Problems 16 and 17. 73 Ma 8, 73 Bh 8, 74 Ma 6, 76 Bh 10
- 3-port directional coupler, S-parameter analysis → §4.5, §3.5. 75 Bh 4
- Directional couplers; four performance parameters → §4.8. 73 Bh 6, 79 Ch 5, 80 Ba 4
- Circulators → §4.9. 70 Bh 5, 72 Ash 5, 73 Ma 5
- Cavity resonators → §4.10. 74 Ma 4, 74 Bh 6
- Gunn diode → §4.11. 79 Bh 4
- MASER → §4.11. 80 Ba 4
- Microstrips → §4.12 and §1.6. 69 Bh 5

## Numerical content

- **This chapter has a numerical companion, Ch4 - RF Microwave Components and Devices - Numerical Problems and Solutions:** rectangular-guide parameters (80 Ba, 77 Ch), cut-off checks (70 Bh, 80 Bh), mode ranking (72 Ma), circular-guide cut-offs (75 Bh), the E-plane, H-plane and magic-tee derivations, the junction problems (77 Ch, 78 Ch, 82 Ba, 82 Bh), the three device identifications and the two radar power problems.

## Syllabus topics never asked in 24 papers

- Termination, phase shifter, attenuators, microwave transistor. Each has a short band above (§4.9, §4.11, §4.12) sized for a 4-mark short note.

## Errors in the sources, corrected here

- Deck slide 18:  $k_y = n\pi/a$  should be  $n\pi/b$ .
- Deck slide 22: figure caption gives the wrong guide size (it is  $5\text{ cm} \times 2\text{ cm}$ ).
- Deck slide 72: Gunn low-field current is not a valence-to-conduction transition.
- Deck slide 77: a MASER is a quantum device, not a semiconductor device.
- Waveguide.pdf p11: TM  $E_\phi$  has a minus sign.
- Paper 81 Bh Q3a: TE<sub>10</sub> is dominant when  $a > b$ , not  $b > a$ .

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short   | Means                                | Short    | Means                                   |
|---------|--------------------------------------|----------|-----------------------------------------|
| RF      | radio frequency                      | MW, M/W  | microwave                               |
| EM, EMR | electromagnetic (radiation)          | TEM      | transverse electromagnetic mode         |
| TE      | transverse electric mode             | TM       | transverse magnetic mode                |
| WG      | waveguide                            | TL       | transmission line                       |
| VSWR    | voltage standing wave ratio          | SWR      | standing wave ratio                     |
| VNA     | vector network analyzer              | SA       | spectrum analyzer                       |
| DUT     | device under test                    | S-params | scattering parameters                   |
| RL      | return loss                          | IL       | insertion loss                          |
| ML      | mismatch loss                        | NF       | noise figure                            |
| BW      | bandwidth                            | SNR      | signal to noise ratio                   |
| Tx, TX  | transmitter                          | Rx, RX   | receiver                                |
| BTS     | base transceiver station             | GSM      | global system for mobile communications |
| EIRP    | effective isotropic radiated power   | ERP      | effective radiated power                |
| SAR     | specific absorption rate             | SARPs    | standards and recommended practices     |
| HERP    | hazards of EM radiation to personnel | RADAR    | radio detection and ranging             |
| ASR     | airport surveillance radar           | CW       | continuous wave                         |

| Short        | Means                                                       | Short   | Means                                                   |
|--------------|-------------------------------------------------------------|---------|---------------------------------------------------------|
| TWT,<br>TWTA | traveling wave tube (amplifier)                             | BWO     | backward wave oscillator                                |
| CFA          | cross-field amplifier                                       | LNA     | low noise amplifier                                     |
| HPA          | high power amplifier                                        | SWS     | slow wave structure                                     |
| O-type       | linear beam tube (axial $B$ -field)                         | M-type  | crossed-field tube ( $E \perp B$ )                      |
| MASER        | microwave amplification by stimulated emission of radiation | LASER   | light amplification by stimulated emission of radiation |
| MESFET       | metal-semiconductor FET                                     | HEMT    | high electron mobility transistor                       |
| GaAs         | gallium arsenide                                            | FET     | field effect transistor                                 |
| BJT          | bipolar junction transistor                                 | PIN     | p-type, intrinsic, n-type diode                         |
| IMPATT       | impact ionization avalanche transit-time                    | TRAPATT | trapped plasma avalanche triggered transit              |
| LPF          | low pass filter                                             | HPF     | high pass filter                                        |
| BPF          | band pass filter                                            | BSF     | band stop filter                                        |
| MAG          | maximum available gain                                      | MSG     | maximum stable gain                                     |
| MIC          | microwave integrated circuit                                | MMIC    | monolithic microwave integrated circuit                 |
| ADS          | advanced design system                                      | IF      | intermediate frequency                                  |
| LO           | local oscillator                                            | RFIC    | radio frequency integrated circuit                      |
| PCB          | printed circuit board                                       | SMA     | subminiature version A connector                        |

| Short    | Means                | Short  | Means                            |
|----------|----------------------|--------|----------------------------------|
| attr(s)  | attribute(s)         | no.    | number                           |
| freq     | frequency            | info   | information                      |
| ops      | operations           | config | configuration                    |
| rm       | remove               | vs, Vs | versus                           |
| w/, w/o  | with, without        | Eg     | example / for example            |
| +Fig     | draw a figure        | (I)    | importance / need / significance |
| →        | leads to / therefore | +      | and                              |
| =        | is / equals          | norm   | normalized                       |
| max, min | maximum, minimum     | calc   | calculation / calculate          |
| eq, eqn  | equation             | approx | approximately                    |
| pts      | points               | coeff  | coefficient                      |
| temp     | temperature          | params | parameters                       |

**Standard and component terms** (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ ) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

**Key mathematical symbols:**  $Z_0$  characteristic impedance ·  $Y_0$  characteristic admittance ·  $\Gamma$  reflection coefficient ·  $S$  scattering matrix ·  $K$  Rollett's stability factor ·  $\Delta$  determinant of S-matrix ·  $\mu$  single parameter stability metric ·  $G_T$  transducer gain ·  $G_{TU}$  unilateral transducer gain ·  $f_c$  cutoff frequency ·  $\lambda_g$  guide wavelength ·  $v_p$  phase velocity ·  $v_g$  group velocity.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

**Tier chips** over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.